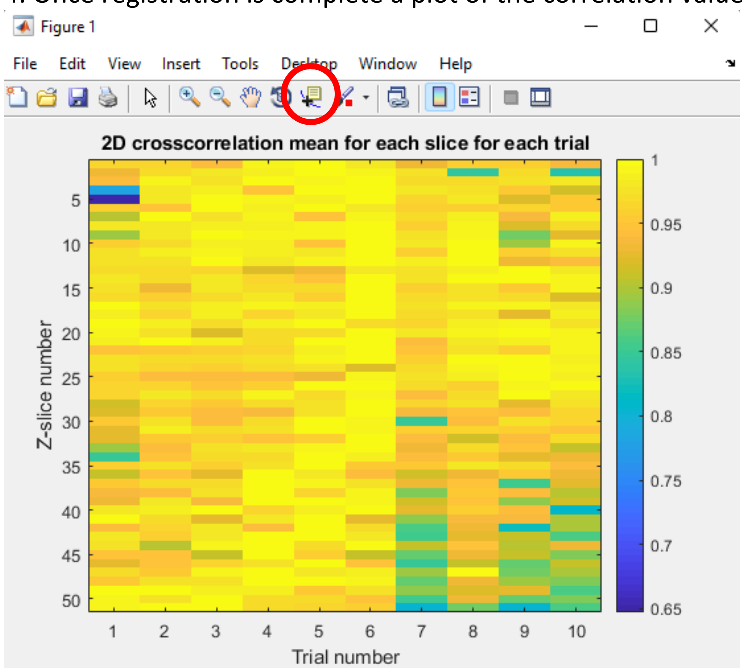
AwakeImgStbl user instructions

Dependencies:
1. Matlab Image Processing Toolbox
2. 'saveastiff' https://www.mathworks.com/matlabcentral/fileexchange/35684-multipage-tiff-stack
3. 'loadtiff' https://github.com/flatironinstitute/CaImAn-MATLAB/blob/master/loadtiff.m
4. Miji, in Matlab "Set Path, Add with Subfolders…", go to your Imagej/Fiji directory and open scripts.
5. Fiji plugin "multistackreg" https://github.com/miura/MultiStackRegistration
6. If using large images, in Matlab goto preferences->Java Heap Memory and increase the value.
7. This code currently only works on PC due to the directory structure.
8. The code is for Bruker/PrairieView file formats using individual TIFF files for each image in a stack.

Data structure:
Input directory with root directory (rootdir) and inside multiple ZSeries folders for each trial with individual TIFF files for each image in the stack.
Output directory, a directory you make to output the analysis, be sure it is outside the input directories.
Put these directory paths in the 'rootdir' and 'outputdir' purple text, being sure to have single quotes.

Running the code:
1. Set reRun = 0 to start with new data.
2. Click Start.
3. Let the code run, Imagej should pop up and start the registration, don't touch and let it run.
4. Once registration is complete a plot of the correlation values will appear.



5. Select the data cursor (red circle above). Click on values to get an idea of a good threshold value.
6. Input the threshold value and hit enter.
7. The code will run, compile the final stack and then do a further registration.
8. If you want to re-run the code using a different threshold, set rerun = 1 and run the code again skipping the image importation steps.

Output:
1. OriginalAverage.tif: The original data averaged without any correction.
2. CorrectedStack.tif: The data after thresholding bit without final registration.
3. CorrectedStack_Registered.tif: The data after thresholding and with final registration.
4. CorrValues.png: Plot of mean correlation values for each trial for each slice.
5. Reconstruction map.png: plot of slice trials included in final image (white) and dropped data (black).

6. RawData\rawSlicexxx.tif: Stack of all trials for a given slice in the z-stack before registration.
7. RawData\registered\rawSlicexxx_reg.tif: Stack of all trials for a given slice in the z-stack after registration.
8. RawData\temp: This directory can be deleted.